

CUSTOMER MILESTONE BRIEF

MS #39 - CICS and VSAM Qualification Hardening

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.39.0
Date	2026-08-30
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #39 - CICS and VSAM Qualification Hardening converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Combines inventory, semantic conformance, and a bounded differential proof for transactional file behavior.

Why this matters to a customer

Combines inventory, semantic conformance, and a bounded differential proof for transactional file behavior.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Bound CICS/VSAM qualification to the canonical graph inventory of 240 CICS commands, 25 transactions, 16 file resources, 15 VSAM clusters, three alternate indexes, three PATHs, and their exact graph identity.
- Added a deterministic 38-case synthetic semantic corpus with 30 targeted boundary cases, 34 passing cases, four explicit fail-closed cases, and 59 observed feature categories.
- Covered KSDS, ESDS, RRDS, alternate-index and PATH lookup, READ/WRITE/REWRITE/DELETE, update tokens, STARTBR/READNEXT/READPREV/ENDBR, file status, RESP/RESP2, ENQ/DEQ, syncpoint commit and rollback, BMS maps, and LINK/XCTL/RETURN.
- Added a 27-entry CICS/VSAM compatibility ledger using all five governing classifications and explicit exclusions for LDS record access, RLS, TSQ, TDQ,

journals, and exits.

- Separated eleven gates for graph inventory, corpus provenance, organizations and keys, alternate indexes, file access, mutation behavior, locking/recovery, CICS command boundaries, exclusions, the existing private account-view proof, and authorized native execution.
- Added frozen schemas, POSIX/PowerShell release-gate integration, deterministic receipts, graph holdouts, exact-corpus checks, negative diagnostics, and rehashed-overclaim tests.
- Kept native VSAM, native CICS, RLS, recovery, runtime, mainframe-equivalence, and production claims false pending authorized region, catalog, concurrency, journal, and recovery evidence.

How it differs from earlier work

MS #39 builds on MS #38 (JCL Qualification Hardening). The earlier milestone established its own bounded capability; this milestone adds cics and vsam qualification hardening while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use cics and vsam qualification hardening as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Kept native VSAM, native CICS, RLS, recovery, runtime, mainframe-equivalence, and production claims false pending authorized region, catalog, concurrency, journal, and recovery evidence.

Source of record

- CHANGELOG.md release section(s): 0.39.0

- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.